

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)		Use of Doppler equation (1) Obtaining data from graph i.e. mean shift = $(4.6 \pm 0.1) \times 10^{-15}$ m OR max and min = $5.55 [\times 10^{-15} \text{ m}]$ and $3.65 [\times 10^{-15} \text{ m}]$ OR max = $5.55 [\times 10^{-15} \text{ m}]$ and $2.54 [\text{m s}^{-1}]$ (1) Amplitude = $(0.95 \pm 0.1) \times 10^{-15} [\text{m}]$ OR velocities = 1.67 and $2.54 [\text{m s}^{-1}]$ OR minimum = $3.65 [\times 10^{-15} \text{ m}]$ and $1.67 [\text{m s}^{-1}]$ (1) All penultimate steps seen or correct values to extra d.p. seen e.g. $\frac{3 \times 10^8 \times 0.95 \times 10^{-15}}{656 \times 10^{-9}}$ and $\frac{3 \times 10^8 \times 4.6 \times 10^{-15}}{656 \times 10^{-9}}$ (1) N.B. obtaining $0.43 [\text{m s}^{-1}]$ or $2.1 [\text{m s}^{-1}]$ gains the 1 st 3 marks	1	1		4	3	
	(b)		Blue and shorter wavelength (or squashed or equivalent or negative shift) Accept negative velocity Don't accept because it is negative	1			1		
	(c)		Use of $v = \frac{2\pi R}{T}$ OR equivalent ($v = \omega R$ and $\omega = \frac{2\pi}{T}$) (1) $T = 390 \pm 10$ days obtained from graph (1) Final sub seen or correct extra dp seen e.g. $R = \frac{0.4 \times 390 \text{ ecf} \times 24 \times 3600}{2\pi}$ OR 2145 km etc (1) N.B. use of this equation is incorrect $d^3 = \frac{GMT^2}{4\pi^2}$		3		3	2	

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				AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)	Use of period equation (1) Algebra (minimum $d^3 = \frac{GMT^2}{4\pi^2}$) (1) Answer: $d = 1.57 \times 10^{11}$ m (1) ecf on T N.B. deriving the equation is ok Don't accept $v^2 = \frac{GM}{r}$ with tiny speed	1	1 1		3	2	
		(ii)	Use of C of M equation N.B. be generous on substitution (1) Mass = 3×10^{25} [kg] (1) ecf	1	1		2	2	
	(e)		1× bad point / something to do with mass e.g. gravity stronger, no idea of atmosphere etc. don't accept gravity between star and planet greater 1× good point / linked to intensity of light e.g. temperature about right, habitable zone, water possible 1× conclusion linked to a point (even if the point isn't quite deserving of the mark e.g. No, it's too hot - 1 mark) e.g. Yes because water possible (this is 2 marks) Accept 2 different conclusions or even sitting on the fence			3	3		
			Question 3 total	4	9	3	16	9	0